----------O(1) - CONSTANT TIME----------

Ex 1: Check Wallet Balance

```
private void btnCheckBalance_Click(object sender, EventArgs e)
{
   int balance = 1500;

   lblBalance.Text = "Balance: ₱" + balance;
}
```

Ex 2: Quick Calculator

```
private void btnCalculate_Click(object sender, EventArgs e)
{
   int a = 10;
   int b = 20;

   int sum = a + b;

   lblAnswer.Text = "Answer: " + sum;
}
```

Ex 3: Pick Your Favorite Fruit

```
private void btnShowFruit_Click(object sender, EventArgs e)
{
   string[] fruits = { "Apple", "Banana", "Mango" };
```

```
    string fruit = fruits[0];


    lblFruit.Text = fruit;
}
```

----------O(log n) - LOGARITHMIC TIME----------

Ex 1: Find a Book Using Binary Search

```
private void btnFindBook_Click(object sender, EventArgs e)
{
    string[] books =
    {
        "Alice in Wonderland",
        "Harry Potter",
        "The Hobbit",
        "The Little Prince",
        "The Odyssey",
        "The Secret Garden",
        "Wonder"
    };

    int left = 0;
    int right = books.Length - 1;
    string target = "The Hobbit";

    while (left <= right)
    {
        int middle = (left + right) / 2;

        int comparison = string.Compare(
            books[middle],
```

```
            target,
            StringComparison.Ordinal);

        if (comparison == 0)
        {
            lblBookResult.Text =
                "Found: " + books[middle];
            return;
        }
        else if (comparison < 0)
        {
            left = middle + 1;
        }
        else
        {
            right = middle - 1;
        }
    }

    lblBookResult.Text = "Book not found.";
}
```

Ex 2: Divide a Number by Two

```
private void btnStart_Click(object sender, EventArgs e)
{
    int n = 16;
```

```
    lblSequence.Text = "";

    while (n > 1)
    {
        n = n / 2;
        lblSequence.Text += n + "   ";
    }
}
```

Ex 3: Number Guessing Helper

```
private void btnStartGuess_Click(object sender, EventArgs e)
{
    low = 1;
    high = 100;

    ShowGuess();
}

private void ShowGuess()
{
    if (low > high)
    {
        lblGuess.Text = "No valid number left!";
        return;
    }
```

```
    guess = low + (high - low) / 2;

    lblGuess.Text = guess.ToString();
}

private void btnTooHigh_Click(object sender, EventArgs e)
{
    high = guess - 1;

    ShowGuess();
}

private void btnTooLow_Click(object sender, EventArgs e)
{
    low = guess + 1;

    ShowGuess();
}

private void btnCorrect_Click(object sender, EventArgs e)
{
    lblGuess.Text = "I guessed it!";
}
```

----------O(n) - LINEAR TIME----------

Ex 1: DNA Results

```
private void btnSeeResults_Click(object sender, EventArgs e)
{
   int[] numbers = { 65, 20, 10, 5 };

   Label[] results =
   {
      lblResult1,
      lblResult2,
      lblResult3,
      lblResult4
   };

   for (int i = 0; i < numbers.Length; i++)
   {
      results[i].Text = numbers[i] + "%";
   }
}
```

Ex 2: Contact Finder

```
private void LoadContacts()
{
   lstContacts.Items.Clear();
```

```
    lstContacts.Items.AddRange(new object[]
    {
        "Ana Santos",
        "Ben Cruz",
        "Carlo Reyes",
        "David Garcia",
        "Ella Flores"
    });
}

private void btnSearchContact_Click(object sender, EventArgs e)
{
    string searchName = txtSearchContact.Text.Trim();

    int checkedContacts = 0;
    bool found = false;

    foreach (string name in lstContacts.Items)
    {
        checkedContacts++;

        if (searchName != "" &&
            name.IndexOf(
                searchName,
                StringComparison.OrdinalIgnoreCase) >= 0)
        {
```

```
            found = true;
            break;
        }
    }

    if (found)
    {
        lblSearchResult.Text =
            "Contact found! Checked: " + checkedContacts;
    }
    else
    {
        lblSearchResult.Text = "Contact not found.";
    }
}
```

Ex 3: Classroom Roll Call

```
private void btnCallJeremy_Click(object sender, EventArgs e)
{
    string[] students =
    {
        "Jeremy", "Ana", "Ben", "Carlo"
    };

    lblJeremy.Text = "";
```

```
    foreach (string student in students)
    {
        if (student == "Jeremy")
        {
            lblJeremy.Text = "PRESENT!";
            break;
        }
    }
}
```

----------O(n^2) - QUADRATIC TIME----------

Ex 1: Outfit Combiner

```
private void btnShowOutfits_Click(object sender, EventArgs e)
{
   string[] shirts =
   {
      "White Shirt",
      "Black Shirt",
      "Blue Shirt"
   };

   string[] pants =
   {
      "Jeans",
      "Shorts",
      "Slacks"
   };

   lstOutfits.Items.Clear();

   foreach (string shirt in shirts)
   {
      foreach (string pant in pants)
      {
         lstOutfits.Items.Add(
```

```
            shirt + " + " + pant);
        }
    }
}
```

Ex 2: Classmate Pairing

```
private void btnMakePairs_Click(object sender, EventArgs e)
{
    string[] students =
    {
        "Ana", "Ben", "Carlo"
    };

    lstPairs.Items.Clear();

    foreach (string student1 in students)
    {
        foreach (string student2 in students)
        {
            lstPairs.Items.Add(
                student1 + " pairs with " + student2);
        }
    }
}
```

Ex 3: Compare Every Fruit Pair

```
    private void btnCompareFruits_Click(object sender, EventArgs e)
    {
        string[] fruits =
        {
            "Apple", "Banana", "Mango"
        };

        lstFruitPairs.Items.Clear();

        foreach (string fruit1 in fruits)
        {
            foreach (string fruit2 in fruits)
            {
                lstFruitPairs.Items.Add(
                    fruit1 + " vs " + fruit2);
            }
        }
    }
  }
}
```